

C1M2_Programming

September 29, 2026

1 Module 2: Programming Assignment

```
[3]: library(testthat)
library(ggplot2)

expect_constant = function(actual, tolerance = .Machine$double.eps^0.5) {
  if (length(actual) == 0) {
    fail("The vector is empty.")
  }

  # Check if all elements are the same within the given tolerance
  if (all(abs(actual - actual[1]) < tolerance)) {
    succeed()
  } else {
    fail("Your answer is not proportional to the posterior.")
  }
}
```

1.1 Problem 1: Inference on the number of trials in a binomial

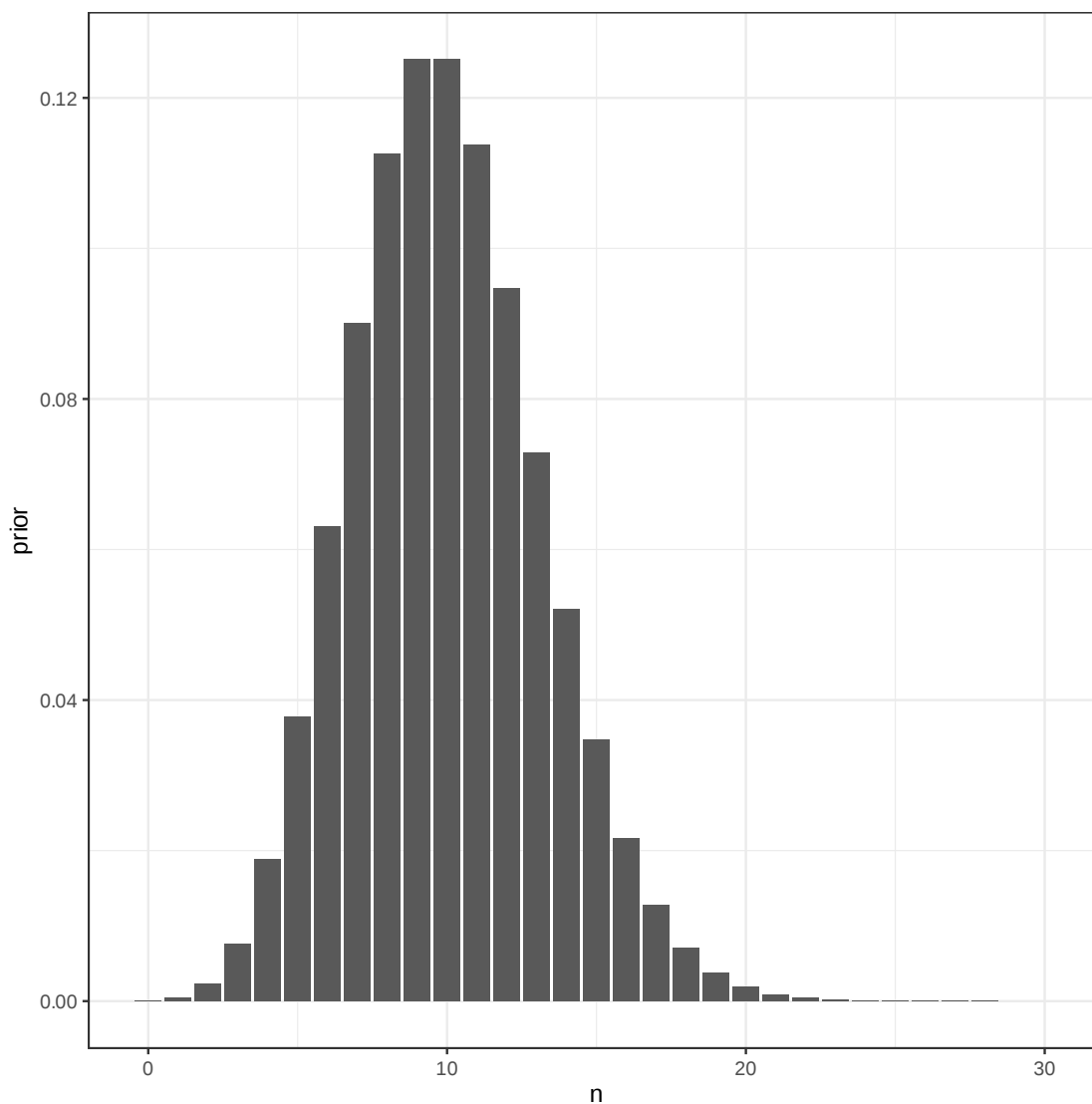
Let n be the unknown number of customers that visit a store on the day of a sale. The number of customers that make a purchase is $Y | n \sim \text{Binomial}(n, \theta)$, where θ is the *known* probability of making a purchase, given that the customer visited the store. The prior is $n \sim \text{Poisson}(10)$.

1 (a) [4 points] Plot the prior distribution for integers $n = 0:30$ using ggplot2. Follow these steps:

- First, using the `dpois()` function, store the values for the prior ($n \sim \text{Poisson}(10)$) in variable `prior`.
- Then, create a dataframe labelled `df` with columns `n` and `prior`.
- Finally, create and ggplot bar plot (using `geom_bar(stat="identity")`) labelled `p`.

```
[4]: n = 0:30
prior = dpois(n, 10)
df = data.frame(n=n, prior=prior)
p = ggplot(df, aes(n, prior)) + geom_bar(stat='identity') + theme_bw()
p
```

```
# your code here
#.NotYetImplemented()
```



[]:

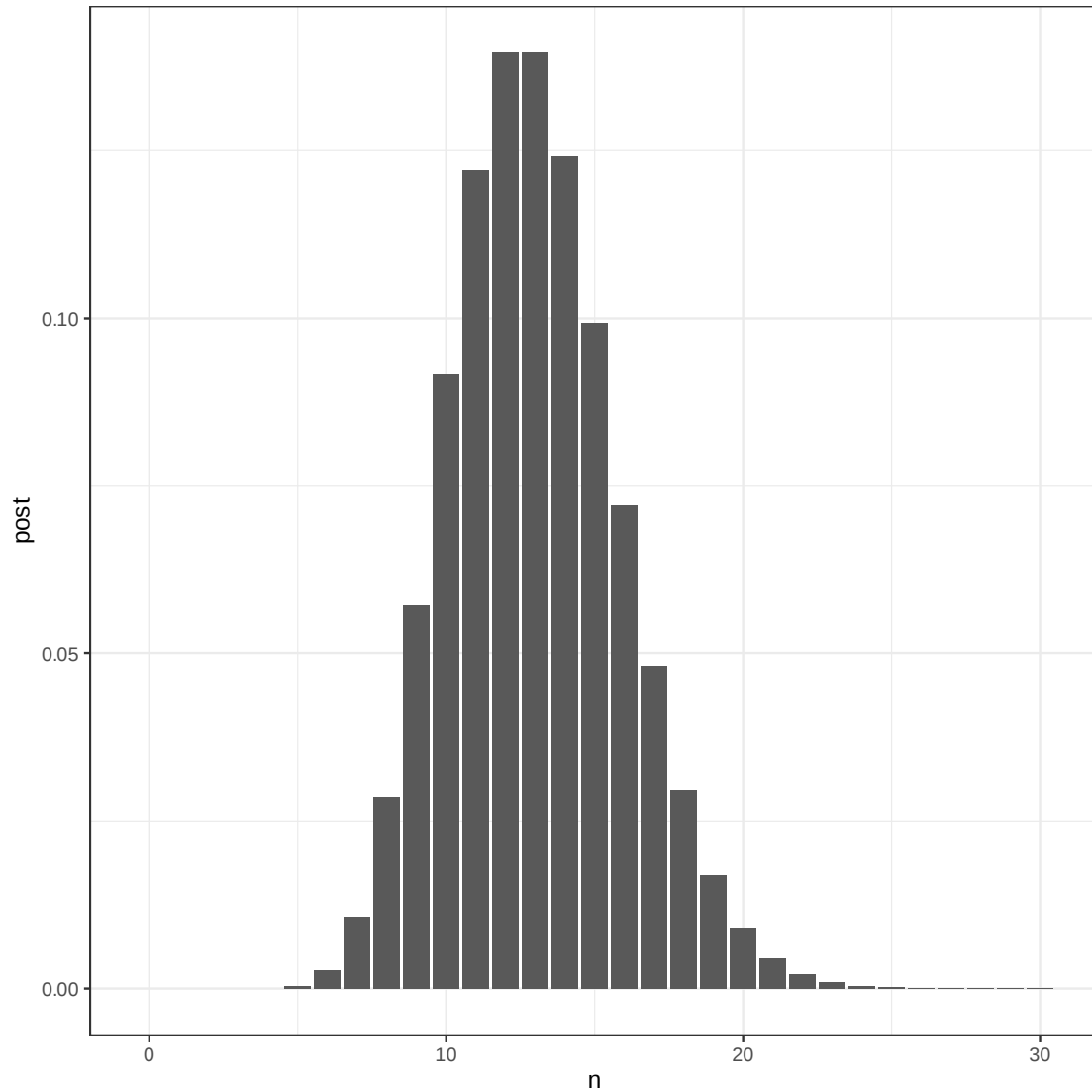
1 (b) [6 points] Assuming $\theta = 0.2$ is known, and that there were $y = 5$ purchases on the day of the sale, compute the posterior distribution for $n | y$ with the following steps:

- Compute a “likelihood vector”, $\mathbf{l}$, where the i^{th} entry ($i = 0, \dots, 30$) of the vector is the probability mass value for the data $y = 5$ and $\mathbf{theta} = 0.2$.
- Compute a “posterior vector”, $\mathbf{posterior1}$, where the i^{th} entry ($i = 0, \dots, 30$) of the vector is the posterior probability mass value for the data $y = 5$ and $\mathbf{theta} = 0.2$. Note that you can easily compute the denominator of Bayes’ theorem using `sum(prior*l)`.

- Plot the posterior distribution.

```
[14]: theta = 0.2
y = 5
#l = choose(n, y)*theta^y*(1-theta)^(n-y)
l <- dbinom(y, size = n, prob = theta)
posterior1 = prior*l / sum(prior*l)
#sum(n*posterior1)
#posterior1
#sum(posterior1)
#df = data.frame(n=n, prior=prior, ll=l, post=posterior1)
#p = ggplot(df) +
#   #geom_line(aes(n, prior, col='prior')) +
#   #geom_line(aes(n, ll, color='likelihood')) +
#   #geom_line(aes(n, post, color='posterior')) +
#   #theme_bw()
df = data.frame(n=n, post=posterior1)
p = ggplot(df, aes(n, post)) +
  geom_bar(stat='identity') +
  theme_bw()
p

# your code here
#.NotYetImplemented()
```



[]:

1 (c) [4 points] Assuming $\theta = 0.2$ is known, but now assuming that there were $y = 10$ purchases on the day of the sale, compute the posterior distribution for $n|y$. Use the same process as given in 1 (b), but store the posterior vector in `posterior2`.

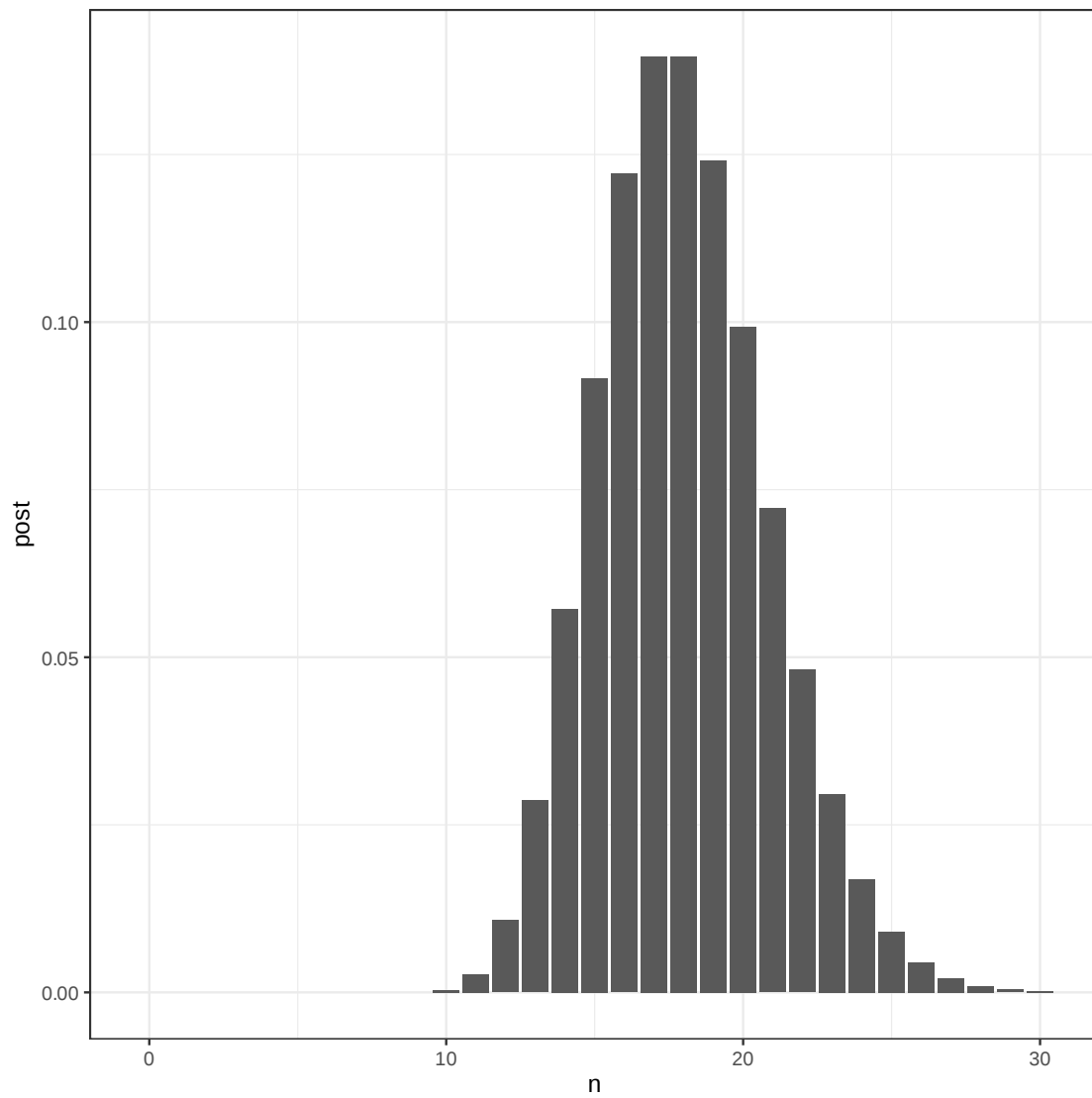
```
[15]: y = 10
      theta = 0.2
      #l = choose(n, y)*theta^y*(1-theta)^(n-y)
      l <- dbinom(y, size = n, prob = theta)
      #print(l)
      posterior2 = prior*l / sum(prior*l)
      #sum(n*posterior2)
```

```

#df = data.frame(n=n, prior=prior, ll=l, post=posterior2)
#p = ggplot(df) +
  #geom_line(aes(n, prior, col='prior')) +
  #geom_line(aes(n, ll, color='likelihood')) +
  #geom_line(aes(n, post, color='posterior')) +
  #theme_bw()
df = data.frame(n=n, prior=prior, ll=l, post=posterior2)
p = ggplot(df, aes(n, post)) +
  geom_bar(stat='identity') +
  theme_bw()
p

# your code here
#.NotYetImplemented()

```



[]:

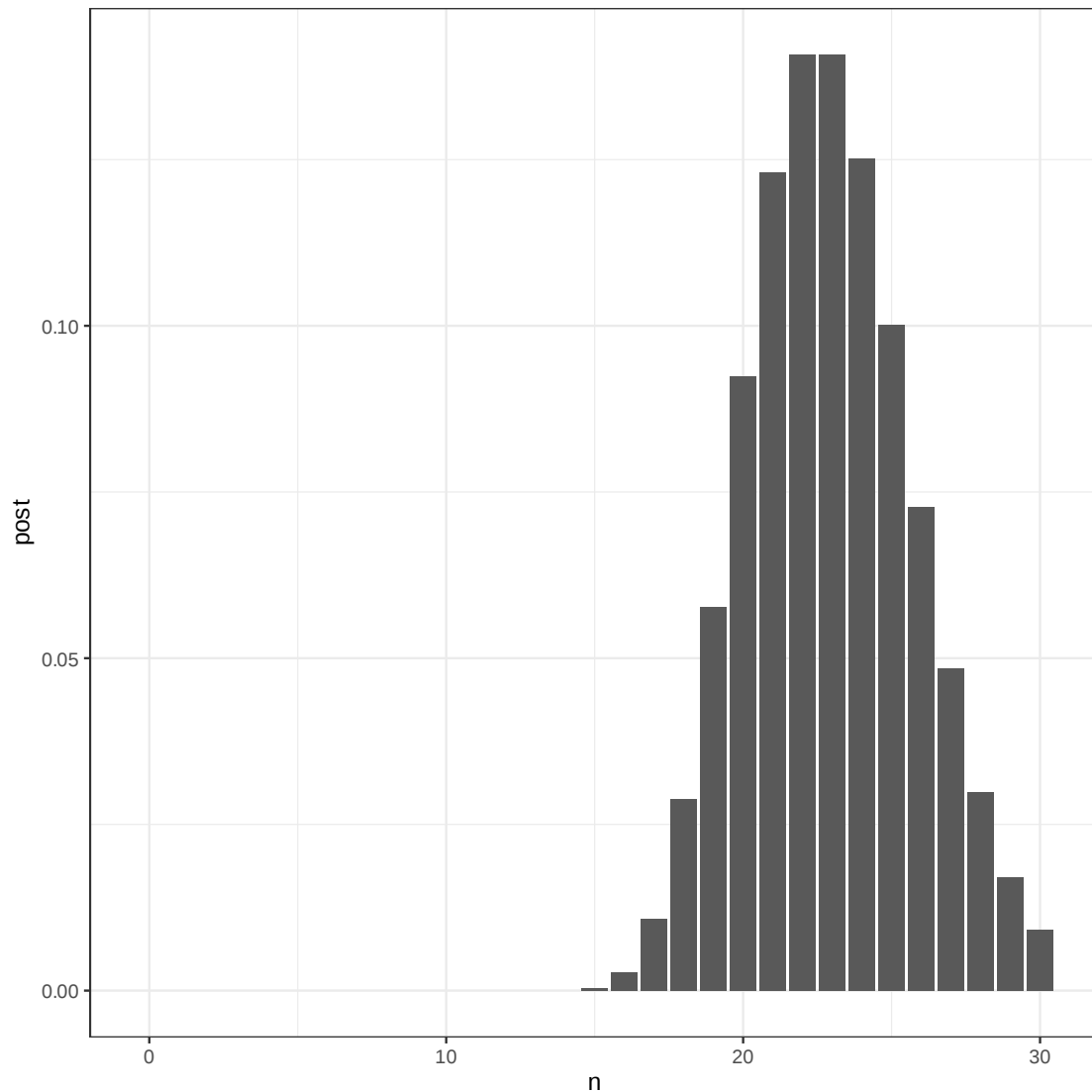
1 (d) [4 points] Again, assuming $\theta = 0.2$ is known, but now assuming that there were $y = 15$ purchases on the day of the sale, compute the posterior distribution for $n|y$.

Start by creating variables `theta` and `y` and assigning these variables with the values 0.2 and 15, respectively. Calculate the likelihood function for $n = 0 : 30$ and store it as a variable `l`. Calculate the posterior and store it as a variable `posterior3`. Finally, using `ggplot2`, create and plot a plot of the posterior labelled `p`. How does the plot of the posterior compare to the plot of the prior distribution? How does it compare to the prior distribution and the posterior from the previous part?

```
[16]: y = 15
theta = 0.2
#l = choose(n, y)*theta^y*(1-theta)^(n-y)
l <- dbinom(y, size = n, prob = theta)
posterior3 = prior*l / sum(prior*l)
#sum(n*posterior3)

#df = data.frame(n=n, prior=prior, ll=l, post=posterior3)
#p = ggplot(df) +
  #geom_line(aes(n, prior, col='prior')) +
  #geom_line(aes(n, ll, color='likelihood')) +
  #geom_line(aes(n, post, color='posterior')) +
  #theme_bw()
df = data.frame(n=n, post=posterior3)
p = ggplot(df, aes(n, post)) +
  geom_bar(stat='identity') +
  theme_bw()
p

# your code here
#.NotYetImplemented()
```



[]:

1 (e) [8 points] Answer TRUE or FALSE for each of the following questions i. - iv. Submit your answer to each by replacing the NA values in the code cell below with your answer to the corresponding question.

- i. When compared with the prior, the mean of the posterior distribution for $\theta = 0.2$, $y = 5$ shifted downward, because the data $y = 5$ suggests that n is likely smaller than was thought under the prior distribution.
- ii. When compared to the prior, the mean of the posterior distribution for $\theta = 0.2$, $y = 10$ shifted upward, because the data $y = 10$ suggests that n is likely larger than was thought under the prior distribution.
- iii. When compared to the posterior for $\theta = 0.2$, $y = 5$, the mean of the posterior distribution

for $\theta = 0.2$, $y = 10$ is larger, because the data $y = 10$ suggests that n is likely larger than the data $y = 5$.

- iv. When compared to the posterior for $\theta = 0.2$, $y = 10$, the mean of the posterior distribution for $\theta = 0.2$, $y = 15$ is smaller, because the data $y = 15$ suggests that n is likely larger than the data $y = 10$.

```
[ ]: i = FALSE
      ii = TRUE
      iii = TRUE
      iv = FALSE

      # your code here
      #.NotYetImplemented()
```

```
[ ]:
```

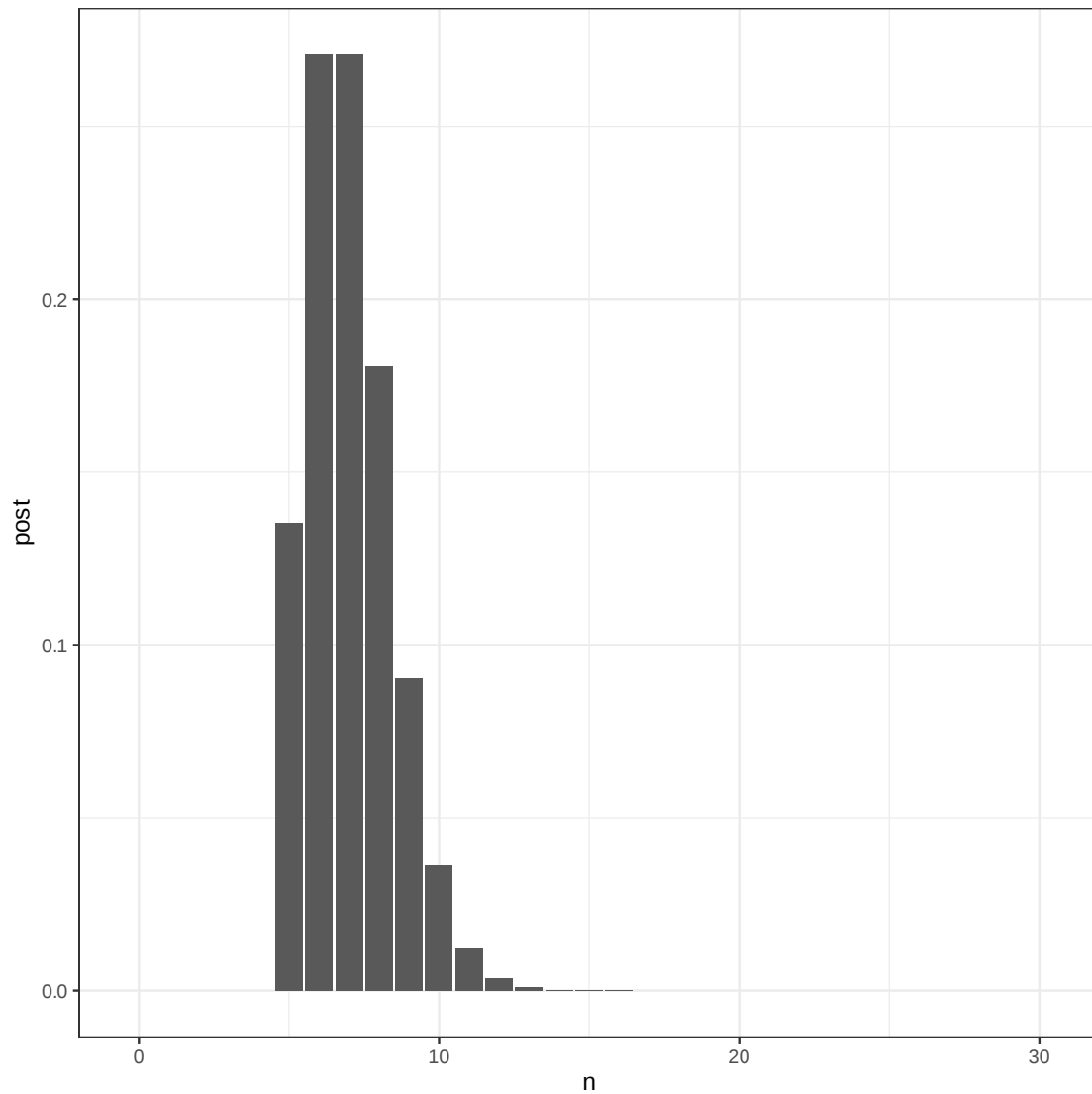
1 (f) [3 points] Repeat part (b) for $\theta = 0.8$.

```
[17]: y = 5
      theta = 0.8
      #l = choose(n, y)*theta^y*(1-theta)^(n-y)
      l <- dbinom(y, size = n, prob = theta)
      posterior4 = prior*l / sum(prior*l)
      #posterior4
      #sum(n*posterior4)

      #df = data.frame(n=n, prior=prior, ll=l, post=posterior4)
      #p = ggplot(df) +
      #   #geom_line(aes(n, prior, col='prior')) +
      #   #geom_line(aes(n, ll, color='likelihood')) +
      #   #geom_line(aes(n, post, color='posterior')) +
      #   #theme_bw()
      df = data.frame(n=n, post=posterior4)
      p = ggplot(df, aes(n, post)) +
          geom_bar(stat='identity') +
          theme_bw()

      p

      # your code here
      #.NotYetImplemented()
```



[]:

1 (g) [3 points] Repeat part (c) for $\theta = 0.8$.

```
[18]: y = 10
theta = 0.8
#l = choose(n, y)*theta^y*(1-theta)^(n-y)
l <- dbinom(y, size = n, prob = theta)
posterior5 = prior*l / sum(prior*l)
#posterior5
#sum(n*posterior5)

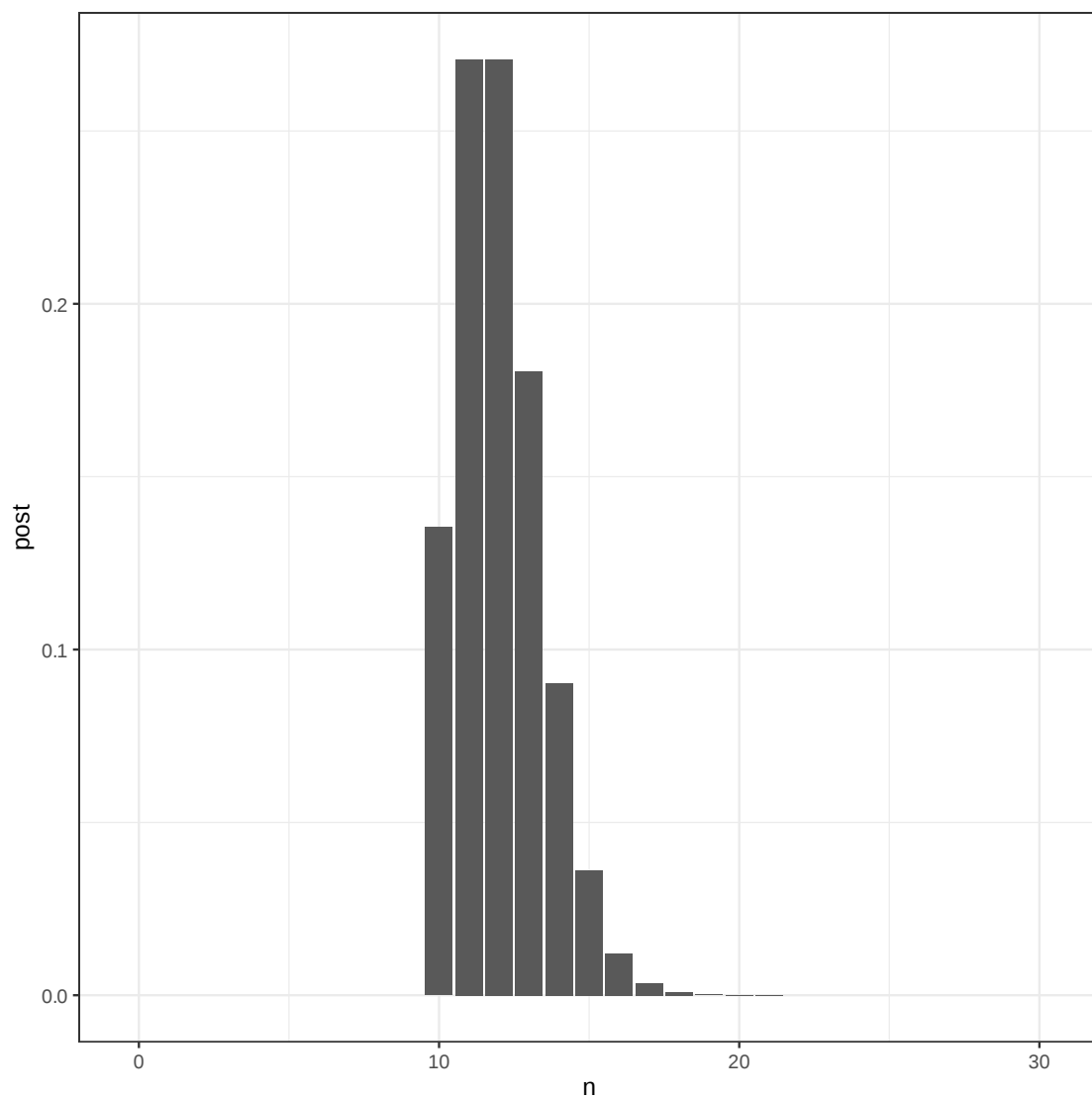
#df = data.frame(n=n, prior=prior, ll=l, post=posterior5)
```

```

#p = ggplot(df) +
  #geom_line(aes(n, prior, col='prior')) +
  #geom_line(aes(n, ll, color='likelihood')) +
  #geom_line(aes(n, post, color='posterior')) +
  #theme_bw()
df = data.frame(n=n, post=posterior5)
p = ggplot(df, aes(n, post)) +
  geom_bar(stat='identity') +
  theme_bw()
p

# your code here
#.NotYetImplemented()

```



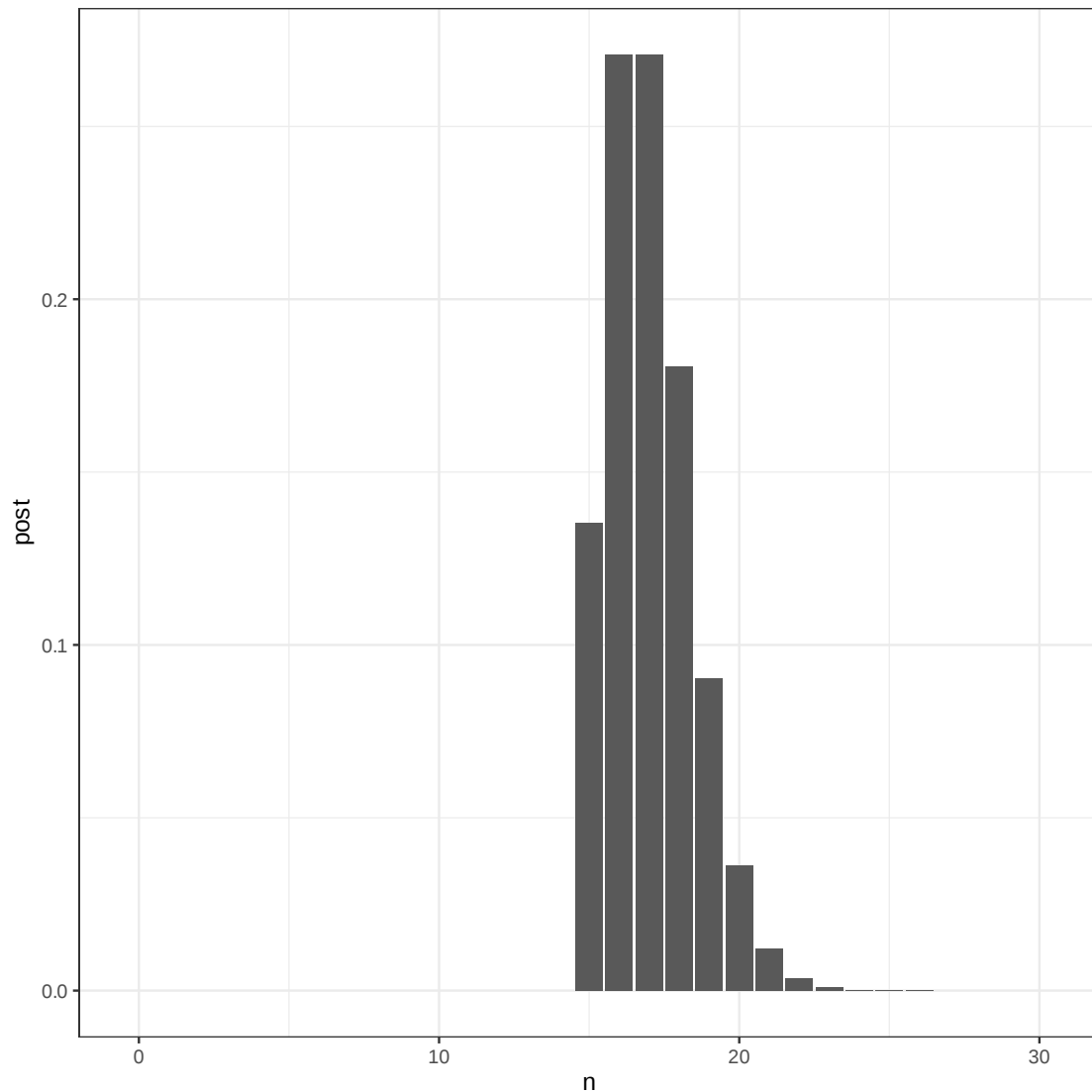
[]:

1 (h) [3 points] Repeat part (d) for $\theta = 0.8$.

```
[19]: y = 15
theta = 0.8
#l = choose(n, y)*theta^y*(1-theta)^(n-y)
l <- dbinom(y, size = n, prob = theta)
posterior6 = prior*l / sum(prior*l)
#posterior6
#sum(n*posterior6)

#df = data.frame(n=n, prior=prior, ll=l, post=posterior6)
#p = ggplot(df) +
#   #geom_line(aes(n, prior, col='prior')) +
#   #geom_line(aes(n, ll, color='likelihood')) +
#   #geom_line(aes(n, post, color='posterior')) +
#   #theme_bw()
df = data.frame(n=n, post=posterior6)
p = ggplot(df, aes(n, post)) +
  geom_bar(stat='identity') +
  theme_bw()
p

# your code here
#.NotYetImplemented()
```



[]:

1 (i) [6 points] Answer TRUE or FALSE for each of the following questions i., ii., and iii. Submit your answer to each by replacing the NA values in the code cell below with your answer to the corresponding question.

- i. When $\theta = 0.8$ and $y = 5$, the posterior distribution has no probability mass above $n \in \{1, 2, 3, 4\}$ because it is not possible for $n \geq 5$ when $\theta = 0.8$, $y = 5$.
- ii. When $\theta = 0.8$ and $y = 15$, the posterior distribution has no probability mass below $n \in \{1, 2, 3, 4\}$ because it is not possible for $n < 15$ when $\theta = 0.8$, $y = 15$.
- iii. When $\theta = 0.8$ and $y = 5$, the posterior distribution mode is approximately 5.

```
[ ]: i = FALSE  
      ii = TRUE  
      iii = FALSE  
  
      # your code here  
      #.NotYetImplemented()
```

```
[ ]:
```